MathArtStream 11 ~ Feedback Welcome

MathArtStream | Feedback Welcome | 1:16:55

All right, hello and welcome.

It is February 5th, 2025, and we're in MathArchStream11.

Feedback welcome.

So, Octopus, if you'd like to say hello, please begin.

Otherwise, we'll pick up where we were just seconds ago.

Yeah, I'd like to say hello. So I'm Octopus. Some of you may know me.

I'm a mathematician, security researcher, and chaos magician.

My different titles and roles that I play at different times.

And I've collaborated with Daniel and Active Inference Institute in the past.

Excited to do so again.

To give you an idea of what we're going for,

this is my current setup.

I have...

I'm on two different accounts in Zoom.

I have two cameras here.

Two web cameras plus the internal camera.

So two accounts, four cameras.

I have two screens that are behind each other.

One is a television screen, one is a laptop screen.

And I have a mirror behind me that may or may not be able to be used.

So I don't have any preconceived patterns of how we use these things,

but I just think it'd be fun to play with different things.

Okay.

We have a few tools, a few links ready.

Got some slides, some Active Inference, possible visualizations,

and some really interesting work that you're involved in with multi-agent modeling.

So please.

I just think...

So, also, there is the YouTube LiveChat Operator panel.

So the LiveChat...

and I'm continuing to rearrange the windows, but it won't always be shared.

I'll just be reorganizing the windows, bringing OBS on, which can kind of always yield this terminating static feedback loop, infinite static.

And also the live chat where I hope we will have interesting feedback welcome.

So with that, and I'm going to be continuing to get the views and everything, so bear with us.

We were paused and began on this idea of feedback and delayed reentry.

And with a question about where and why and how is audio feedback different than visual feedback?

So maybe where does that take you and I'll put up what I can.

Yeah, so a couple of things that brings up for me.

One is that I've done those experiments.

I've done those experiences since.

And right now I'm doing that experience because I'm hearing the audio feedback from the phone.

But it is extremely disorienting.

So if you see the video feedback, you can probably hear the audio feedback from my phone as I'm hearing the echo on myself.

And I find it very, very difficult.

Like I couldn't, I could not produce a coherent stream of thought in this type of environment.

But my strategy right now is to delay so that there's a chance for the feedback to die out.

And I'm not doing it directly in my ear, which I think would also greatly stuff.

But the strategy that I found that I can use sometimes and have used in the past when I did similar experience was to speak very slowly so that it almost sounded like the feedback I was receiving was occurring at a clearer delay from the signal that I was generating.

So when I speak very slowly like this, I don't find it as disorienting.

And in fact, I tried to record a song this way back when I was doing recording music as something I was pursuing.

And I found that it produced something of a very interesting effect.

And the other thing that I found helpful was to intentionally adjust my timbre so that the body resonance gave me a different kind of signal beyond what I was hearing on the audio itself.

But I agree with you that it is extremely disorienting.

However, I've, I've heard some different perspectives on that.

And one of the perspectives is that if you're playing an instrument, it's actually a very intriguing experiment to hook a synthesizer up to an audio output and to put it through a delay so that you're hearing something different than what you were, like what you're playing at that moment is not what you're hearing.

You're hearing what you're hearing, you're hearing what you played a few seconds ago, then people who do this often find based on what I remember being told, I need to find sources for this.

But that that can feel very natural and freeing because it lets you out like it lets the critic out of your feedback loop of your playing.

So I think that the experience of time and the fact that I wonder if it's connected to the fact that audio hallucinations are so common.

Because it feels to me like there is a hallucinatory effect that we're having to fight through and we're having to fight through our, like our, our cognitive systems interest in, in not being fooled by hallucinations.

So visual feedback is kind of more interesting, but it doesn't some, I don't think I'm visually generating in the same way that I am.

ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail ail

ail ail

through the same sort of circuitry,

like they're very close physically.

And so the belief that I'm hearing voices

that aren't mine is a little disorienting.

But visually, when I see visual feedback,

I know there's something artificial about that.

And again, I'm not speaking as a neuroscientist.

Let's not say I'm an expert in any of this,

but I don't have any,

I guess I'm experienced enough as a psychonaut

to have some experience

and to have some read some things about it

from a non-scientific, more experiential perspective.

One of the first things we look for in signs of insanity

is the idea that the person is hearing voices, right?

So I also think there's a civilizational aspect of it

where if you're hearing voices,

is that something you're taught not to do?

Or you know that there's the potential feedback

of social judgment

and that that can have negative consequences?

I don't know if that's connected at all

to the physiological and neurological aspects

of the experience,

but it's another kind of interesting perspective to consider

because I think ultimately what feedback is giving us

is a different perspective

on an experience that we've recently had.

So it's not disorienting

if we're hearing a tape of our own speech, right?

If you put on a recording of me saying this

and I had to talk over it,

I would not find that difficult.

So I think it has to be about the actual physiological aspect

of the circuitry involved
and something about the time delay and distortion aspect.
There's, this can be used, right?
This can be used for, for instance,
electronic music production.
There are algorithms that intentionally create
these feedback loops
for synthesizing the sounds of stringed instruments.
I learned about this, I think,
in Julius O. Smith's book on signals and filters
because this is like feedback loops are useful
in design of filters.
I think it's called the Strong Carplus algorithm
and K-A-R-P-L-U-S.
And the whole point is that to,
to make a, an excited sound,
like a, a violin that's string that's being excited
or a string that's being plucked,
you give it a very brief initial signal
and then use a delay line
to let the feedback create the synthesized sound.
So I think the audio feedback is also different
when it's not, when there's not such a strong
semantic or semiotic aspect to it
where we're not trying to make sense
of what we're saying.
If I were to, here's an interesting experiment.
I will speak nonsense syllables
and see if I find it difficult
to generate those extemporaneously.
Lo so, lo so, lo so, lo so, lo so, lo so, lo so, lo so, lo so, lo so, lo so, lo so, lo so, lo so.
See, I didn't feel anything difficult about that.
And it could be open-label placebo
where I didn't, you know,
it's self-fulfilling prophecy.
But that makes me wonder
if the difficulty
and the disorientation comes from

the fact that typically
we're working with words
and so there's a meaning aspect there
expecting it to mean something.
What do you think, Daniel?
I welcome your feedback.
Yeah, thank you, Octopus.
Well, you began with this
directly empirically verifiable
and also I'm sure many neuroscience papers
on which kinds of feedback re-entry
for audio
stymie natural speech.
And then you talked about
different ways to change the rhythm
and that's kind of like
a different feedback producing setup.
So visually like having this
with OBS
but because Zoom
with its special above OBS even
so here even
but if Zoom weren't
if I were in a different
screen share mode
then
or if I pull the Zoom off
to the other screen
then there's the fixed.
So here with the frame rate
and then with OBS
with the video settings
I don't think the frame rate
can be changed
during the stream.
but a lot of configurable video settings
and then this kind of comes
to those

different kinds of feedback
just to even set some of the
possible types
that are really important
for the generalized cybernetics
that I'm sure will be
interesting to hear about
and you can share your screen too.
We have different sensory interfaces
visual and audio
everything is through time
in that sense
OBS too
but again that
way in which
the visual interface
possibly feeling at a distance
compared with sound
which is kind of an ongoing
probably more approximate
event
and just the controllability
of the visual
and the putting it in a box
on our computers
and then that
opens up this
well there's a live chat
so if we're thinking about feedback
we want a generic definition
of feedback
agent
thing
all these different
research directions
on generalized
dynamical systems
generalized cybernetics

all these kinds of things
how do we
scope and survey
and lay out
where are we dealing with
a 200 millisecond
speech delay
with the audio
and then
sound and noise
how do we reconcile
that
gripping
and symbolic
nature
of both
audio and visual
as to
available
streams
but not
exhausting
this bigger
research
question
of generalized
feedback systems
and a ton of
practicalities
about
feedback
and reentrant
loops
in
any kind
of
offline
or purely

online
system
so I'd like to
propose
an angle
here
that's
probably not
what might
traditionally be
expected
but I think
is very relevant
especially for like
management cybernetics
and
we're learning
which is
the
construction
of grades
in a pedagogical
setting
specifically
the construction
of
student work
that is then
given
instructor feedback
and
that this is
something that I
have
experience with
and I think
it might be a good
jumping off point

for discussion
of other
types of
feedback systems
because it
doesn't presuppose
any knowledge
of engineering
and
I thought you
were going to
just copy paste
the same thing
down the screen
and I was about
to think
that up
so
so
this is actually
something that
undid me
in teaching
on zoom
is I
got way
too much
into
the
like
the specifics
of rectangles
and the idea
that you can make
all these different
rectangles appear
on the screen
but

the
the grading
aspect of this
is really important
because you know
there's this
interactive process
at some time
scale
one of the things
I learned
when I was teaching
calculus
was you know
students were
understandably
concerned about
their grades
but if I gave
any type of
automatic feedback
system
to reduce
work for
myself
that could
create an
emotional feedback
loop
so the example
here is
something like
web work
where
you can go
in and
this
the instructor

can assign
questions
as a homework
set
and there are
many other systems
that do the same
thing
but then
you know
the student
types in
a
student
types in
their
answer
and gets
immediate
correct or
incorrect feedback
well this could
produce a feedback
loop where
a distressed
student on a
short time frame
keeps getting back
the answer
and as they type in
the answer and get it
wrong they become more
and more frustrated
as they become more
and more frustrated
their capacity for
rational thought
decreases

and a sense of
desperation
kicks in
as a sense of
desperation
increases
and they can
they're worse
at doing math
in this frame of
mind
they now
are more likely
to get the answer
incorrect
and now
suddenly
without any
limit on the
number of
attempts
they can just
go
forever
and it can
actually be a
very negative
emotional spiral
to get kind of
trapped in a
question
close to
midnight
where you've
waited a while
and
so that's
that type of

spiral feedback
loop of
reinforcing
negative emotion
less capable
logical thought
incorrect answer
negative feedback
in terms of
you did not get
this correct
was one of the
first feedback loops
that I think I
really understood
and needed to
take action in
some way
to
stop essentially
because I
did not want
to create
a classroom
where all the
students have
been up late
the night before
so the
different instructors
have different
strategies for
this
one is
you can
be available
whatever you
choose

to make the
deadline
close to
so for
instance
you could
you know
you don't go to
bed till late
if the deadline
is midnight
or you set
the deadline
at 2pm
when you're
going to be
around to
give last
minute help
or you set
the number
of attempts
you don't let
students have
unlimited
attempts
or you create
like a practice
set that has
unlimited attempts
and then a
graded set
that has
you know
three attempts
because the
other type of
feedback is that

students can
just begin to
engage in rampant
guess and check
and you don't
really want that
either right
that they just
start typing in
one two three
four five six
start guessing
numbers and
taking advantage
of the unlimited
attempts
then the third
type of feedback
is to introduce
another participant
which students
would often do
which is to just
take the question
to Wolfram
Alpha or
ChatGPT
which is likely
to give them
the correct
answer if
they become
skilled at that
process
but then at
that point
the calculus
class just

becomes about
translation
the student's
skill at
translating
between two
artificial
intelligences
and while
that might
be a
direction that
humans need
to go
that's not
where you
want the
primary rewards
in calculus
class to be
that the best
grades are those
who have
given the
most thought
to how
to get
Wolfram
Alpha to
answer
WebWorks
questions
and it
doesn't
the precise
computational
engines here
are not too

important
it's more
about how
do you create
a system
where the
rewards
come from
the activity
that you're
actually hoping
to reward
and are not
easily
gameable
and so
what's needed
is some
type of
control
something
that
prevents
the infinite
feedback loop
from happening
something
that stops
it
something
that
lessens
some part
of the
amplification
I see
there's
something

here
so true
about
feedback
and
education
we can
really
influence
approaches
to
studying
bring it
in a
different
way
yeah
going into
the browser
bar
make it
plain text
make it
into a
string
reward
shaping
yeah
a lot
to say
about
the
landscape
of
epistemic
and
pragmatic
value

in terms
of how
we think
about that
in active
inference
and expected
free
energy
and then
where is
there a
stepwise
change
in
epistemic
and or
pragmatic
value
like a
well
opening
in the
middle
of a
flat
plane
and then
where is
there
so-called
dense
reward
function
with a
broad
basin
of a

attraction
and a
lot of
learnable
targets
not just
rugged
stabs in
the dark
and so
that's
part of
this
broader
motivation
take the
challenging
approximation
and relate
it to
a
tractable
incrementally
optimizable
procedure
or operation
what do you
think about that
first part
sorry
can you say
I didn't
capture all
of it
like there's
some statistical
landscapes
where the

ball rolls
downhill
convex
optimization
star
anywhere
and following
the local
gradient
is going
to be
maximally
efficacious
to find a
global
optimum
and then
contrast that
with kind of
the other
extreme
setting
high
dimensional
flatness
no local
gradient
or signposting
and then just
there's one
point but even
chaotically close
to it
you could end up
infinitely far
away
which is a
desired property

for cryptographic
algorithms
where you want
like the
next
nonce
or next
piece of
salt
to take
you to
undetectable
place
in a space
and then we
have this
optimization
characteristic
with the
gradients
in the way
that local
free energy
or statistical
learning
or loss
function
reward
learning
or more
energy
based
that local
optimization
and it's
like
that's cool
and interesting

and we
can go
into it
it's a
practical
question
when doing
machine
learning
and statistics
so there
are actually
two questions
there I think
the first
is thinking
about the
fitness
landscape
and where
agents will
find rewards
but kind
of in
physical
systems
there's
in physical
systems
we don't
like in
an engineering
physical
system
there isn't
initially
at least
I don't

think
it's important
to think
about the
danger
of simulation
so if you
have a signal
that you want
to control
you want to
filter out
part of it
the component
is not active
and can't
try to fool
you into
setting
parameter
settings
that will
then
make it
so that
it can
let the
undesired
signal
through
later
whereas if
you're dealing
with active
agents
like college
students
or potentially

AI agents
or humans
interacting
with AI
systems
we do
have the
ability
to think
through
how
simulating
something
initially
might lead
to a
response
which then
can
the response
itself
can be
exploited
like a
trap
essentially
so physics
doesn't
set traps
at least
simple
physical
signals
like a
sine wave
isn't going
to set a
trap

it doesn't
have that
type of
cognition
or strategic
thinking
but a
college student
does
a college
student
can
optimize
for
if what
really
matters
is the
grade
and not
the
learning
how can
I
optimize
the grade
with minimal
learning
with the
understanding
that then
there's a
snowball
effect
getting an
A in
this class
increases

my social
status
among
the gossip
network
of the
faculty
it
will
improve
my GPA
which is
like a
terminal
metric
essentially
and
the
fitness
landscape
for
learning
is also
made
difficult
by the
fact
that the
epistemic
feedback
of how
do I
know
that you
know
what I
want you
to know

already
there's a
feedback
loop
that I
have to
kind of
I don't
know
it made
me crazy
so I
stepped out
of doing
it for a
bit
but still
the thoughts
I think
are valuable
and they're
like these
patterns
that we
could look
for
yeah
two
thoughts
on that
one
this is
a
inverted
image
color
on a
usually

white

pdf

however

it's

showing

this

particular

kinds

path

intervals

particular

kinds

and strange

things

strange

furry

things

like

needing

to

accept

cookies

and

it coming

up in the

textbook group

which

you did

awesome

and fun

math work

for

and that

question

of

from

sine wave

or

just like
y equals
two
it's a
number
it's also
a function
through
time
through
time
for
all
time
steps
do
too
so
those
sorts
of
pedagogical
where
the
essences
and the
patterns
can sometimes
be
distilled
and
fractionated
on through
more sophisticated
strange
cognitive
modeling
and all

that
entails
and then
also
the
I think
it's a
it's an
apocryphal
story
but some
okay so
we're going to
play a game
everybody's going to
send a letter to
the editor
picking a number
between
zero and
a hundred
and then
you want to
guess
the number
that's two
thirds of the
average of all
other
people's
choices
so it's
like if
everybody
picks a
hundred
I could
take

67
but if
but then
people will
definitely
take 67
then
so then
all
guess
43
and then
oh everyone's
going to do
that though
so then
where do
people
bottom out
even in
that very
reduced
setting
with
would people
really pick
like
0.0000002
and then
if they would
really do that
why wouldn't
I also
go lower
and then
people pick
like 17
or you

know
they pick
some
finite
number
even
though
the
game
theory
kind of
open-ended
simplified
game theory
might have
these mathematical
properties
like
convergence
of sampling
performance
on a test
or a given
training data
set
all these
different
features
but
just because
they're one
way
then the
causal
structure
of the
world
could

change
and that
could be
even better
anticipated
by the
causal
model
of the
agent
or could
just
absolutely
flip the
signal
orthogonally
and just
reduce the
efficacy of
that model
low
I like
the simplicity
of that
game
I think
that's one
of the
difficulties
in these
types of
very complex
systems
is if
if all
you can
study
is something

with
overwhelming
intricacy
and complexity
then you
basically have
no hope
you need
something to
stay stable
or repeatable
and
of course
there is
the
the idea
that
in psychology
people who
are aware
of certain
research effects
become immune
to the
impact of
that effect
like if
you
tell people
the expected
outcome of
an experiment
they may
take that
information
and
they may
take that

information
and then be
immune to
the experiment
essentially
because
they know
that they're
not supposed
to become
prison guards
or they're
horrified at
the thought
that people
could become
you know
induced by
authority
to engage
in sadistic
behavior
against a
stranger
so that's
another thing
is there's
this metagame
aspect
of playing
a game
but you see
how other
people have
played the
game
the game
itself evolves

over time
often in a
co-evolutionary
way or a
cyclic
evolutionary
way
you see
this in
professional
football
the national
football
league
there's a
movement
towards
there was a
movement
towards
throwing the
ball more
with lighter
faster players
and being
more mobile
so defenses
adjusted for
that and
made their
players also
lighter and
faster and
more mobile
well now the
counter evolution
on offense
is to go

back to
heavier
stronger players
and force
the ball
over these
lighter stronger
defenders with
a run action
rather than
because as they
have evolved
they have evolved
to a weak
state which you
can then swing
back and
exploit
as we see
the game
repeatedly
played
we learn
something about
how to play
it
what we learn
about how to
play it may
or may not
ultimately make
us better
players or it
may just become
another level
of the game
where signaling
a specific

strategy can
be part of
the strategy
itself which
may not be
the actual
strategy
this is the
first figure
from the
2022
active textbook
and
we could even
unpack why
an arrow
is graphically
how it is
and chase
that
however
the
action
perception
interface
question
is often
called a
loop
and
certainly
from a
program
perspective
like a
Turing tape
perspective
not that

that's
everything
but from
a
procedural
engineer's
perspective
there can
be like
two
ongoing
processes
but on a
single thread
it gets
bottlenecked
to like
something
happening
procedurally
and so
it's so
interesting
how the
instantaneous
nature
of the
organism
environment
interface
like
there's no
feedback
from
this
patch of
skin
to

that
patch of
skin
a month
later
but yet
a model
making a
measurement
from that
location
might be
able to
have an
arrow
and a
number
that we
call
feedback
from the
math
perspective
like this
kind of
causal
gravity
that flows
around the
horn
and has
a kind
of
statistical
force
and we
call it
a feedback

loop
like
others
there's a
feedback
loop
between
the
supply
and the
demand
or the
Laca
Volterra
ecology
models
those are
also called
feedback
cycles
but
they're
instantaneous
all along
the way
they don't
necessarily
see
11 time
steps
ago
with an
arrow
influencing
them in
the moment
and
outside of

time weaving
and looping
and all
that
it doesn't
even necessarily
happen
that's
kind
of
the
Markov
Markovian
property
like
past
only
influences
the
future
through
the
present
but
if
everything
is
passed
through
the
constraint
of the
present
then it's
like
huh
that is
interesting

there are
these
feedback
loops
it's
also
true
that
an
objectively
incorrect
belief
about
the
world
can
be
advantageous
as a
strategic
choice
this is
basically
hyperstition
this is
also
why
you know
historical
the
constant
renegotiation
of history
is really
powerful
because we
kind of
step into

this world
that has
this history
of how
the game
was played
and part
of our
strategy
for the
game
is based
on the
outcome
of previous
games
so
when we
come to
some
dispute
about
how
games
were
played
in
the
past
and
what
the
outcome
was
and
the
interpretation
of that

it has
a powerful
influence
on how
we currently
play the
game
could you
share a bit
about this
series
and where
is it
fitting in
today with
your broader
research in
this and
what is this
sequence of
workshops
get at
specifically
sure
and after
we do
that
we wound
up in a
very kind
of
sober
focused
and
intellectual
place
which I'm
happy to

be in
but after
we do
the discussion
of that
research
I was
wondering
if we
could go
back to
focused
and free
exploration
of the
actual
patterns
as they
exist
I'm not
making an
explicit deal
but I'm
just kind
of
setting for
myself
on
what we
can
have
free
feedback
yeah
free
feedback
free
feedback

hyperstition
planning
planning
is a type
of feedback
right
because then
if we
plan
we now
have
something
to have
feedback
about
over
behind
schedule
for instance
I mean
the schedule
was arbitrary
and made
with no
cognition
behind it
but still
we now
have a
schedule
which is
telling us
where we
were supposed
to be
so now
we can
the setting

of expectations
is a
form of
feedback
it's a
past belief
about
where you
should be
now
and then
that can
actually be
very powerful
in the
form of
but yeah
I'm happy
to talk
about this
research
and so
this is a
series that
I did
working with
block science
originally
I was
working off
of some
notes written
by Dr.
Michael
Zargum
the CEO
and senior
engineer

at block
science
on this
idea
of
research
he had
done
into a
type
of
dynamical
system
called
generalized
dynamical
systems
and
essentially
dynamical
systems
usually
are
advanced
by the
application
of a
single
specific
rule
so for
instance
the
logistic
map
where you
just take
x sub n

plus one
and it's
equal to
alpha
times
x sub n
times one
minus x
sub n
all in
parentheses
they advanced
by a
singular rule
whereas in
generalized
dynamical
systems
these were
created
initially
with the
idea
that you
were dealing
with systems
where there
would be
some type
of switch
or some
type of
input
at a given
point
and so you
needed a way
where not only

were you applying
a single
rule
but you
were actually
responding to
actions that
were taken
within the
system
and naturally
that sort
of branches
out into
the idea
that there
need to
be agents
that the
system
now has
components
that have
agency
so it's
moving from
a purely
deterministic
dynamical
system
where the
components
proceed
according to
a completely
pre-described
mechanism
and moving

towards
a system
where the
components
have agency
and that
naturally
intersects
with the
idea
of agents
so
agents
are
you know
multi-agent
systems are
a very
vast area
of research
that are
especially
important
in economics
and biology
and other
places
any type
of social
system
or agents
interact
I came
to them
via
cellular
automata
but again

the agents
there are
cells
who are
engaged
in behavior
that's
mechanistic
the cells
in a cellular
automaton
don't have
individual choice
they follow
the same
rule
and yet
you still
see emergent
behavior
so when
you add
in agency
and the
type of
feedback
loops
that are
emphasized
in active
inference
now you're
going to
see
the possibility
of different
types of
emergence

based on
the
choices
that the
agents
make
and so
there's
a formalism
there's a
there's a
computational
tool
called
CAD-CAD
or
computer
aided
design
of complex
adaptive
dynamics
which is
working with
the CAD
framework
from engineering
where you're
using
computer aided
design
to
work with
mechanical
systems
but now
you're using
computer aided

design to
look at
complex
systems
according to
a well-defined
syntax
that lets you
define what
happens at
different stages
of an
interaction
and this
connects to
a type of
economic
model called
multi-agent
influence
diagrams
which are
a type of
causal
inference
structure
in
economics
that were
introduced by
Culler and
Milch
and
all three
of these
there are
maps between
these things

I think of
these as a
mathematician
sometimes I
think in
terms of
category theory
or categories
for me it
was like
there are
obvious
morphisms
between
these
categories
that
they're not
three distinct
things that
are just
because it's
not too
hard to
translate
between the
three different
realizations
of a system
and so
I was
exploring that
and playing
with it
it improved
my understanding
of CAD-CAD a
lot I

appreciated
the elegance
of CAD-CAD a
lot more
as a tool
after
kind of
translating it
into terms
that my brain
was more
comfortable with
but yeah
it's a good
series of
workshops
I enjoyed
doing it
and I
don't think
I'm happy
to get
feedback on
from people
and engage
in conversations
with people
about
I think
the most
interesting
thing for
me
is that
as soon
as we
now have
structure

we can
gain
understanding
through that
structure
so the
point of
introducing
this notation
is not
to just
look more
formal
but if we
have
something
and we
say oh
this is a
directed graph
this is a
labeled
directed
graph
like
Dr.
Zargum
drew that
image
and so
he has a
labeled
directed
graph
with
subgraph
components
well now

there are
lots of
systems that
have
labeled
directed
graph
with subgraph
!
components
and we
know how
to do
things like
add an
edge
add a
node
change
a
label
change
we know
these
different
structures
and how
we can
study
them
with
these
parameters
like
families
of
structures
right

and that's
kind of
where my
brain is
drawn
my
neurotype
is such
that I
tend to
get
overwhelmed
with
immense
complexity
at once
and personally
the research
journeys where
I've been
the most
successful
were the
ones where
I could
find a
starting
place
and identify
some
specific
parameters
to kind
of tune
the knobs
on
and so
like giving

this type
of problem
this type
of structure
gives us
an entry
point
where
from my
perspective
now not
only do we
understand it
better and
translate it
to ourselves
better
and can
communicate
it with
interested
non-technicians
better
and communicate
it amongst
ourselves
better
but also
we can look
at specific
aspects of
the structure
to see
how specific
structures
behave
in ways that
we might be

able to
predict
and this
is the
starting point
of a lot
of the
complex
systems
research
for instance
the game
of life
when John
Conway was
designing the
game of life
cellular automaton
he tried
several different
rules
just guessing
and checking
to see which
ones might
have
which parameter
values for
birth and
death
might lead
to interesting
configurations
now of course
we can do
that in a
computer
so the

possibility of
searching a
parameter space
is greatly
amplified
but in order to
be able to
search a parameter
space we have
to have an idea
of what the
parameters to
the system
are and
then how we
will refine
our parameter
search so that
is again a
feedback system
where we
generate we
know the
parameters we
want to work
with we
generate them
we generate the
system performance
we assess the
performance we
can then go
back and
look at
whether like
what parameter
settings might
be useful

gradient descent
is a feedback
system in this
way
and there are
lots of search
and optimization
procedures
which are
basically
generate and
test it's
just a matter
of how the
potential solutions
are generated
and how they're
tested and then
the testing
hopefully hooks
back up to the
generation so
that we have a
better idea
of where to
look in the
search space
and I realize
that all of
this is probably
well familiar
to the viewers
here who are
familiar with the
active inference
framework like
they're familiar
with lots of

different systems
that involve
feedback loops
and modifying
behavior based
on feedback loops
but for me it's
always helpful to
just go back to
the beginning
and see the
basic patterns
as a starting
point
awesome
in our last
descent
which might be
still
somewhat extended
first if anyone
has a
feedback
comment
they can go
for it
optimization
as feedback
or optimization
in feedback
whether it's
the OODA
loop
or the
TOTE
test
operate
test

exit
all these
different
guess and
check
methods
and you
brought up
planning
earlier
this is
something
that's
come up
in the
philosophical
and applied
literature
where is there
a plan
where does it
look like
from another's
perspective
it could also
be modeled
with a plan
and then
nested
planning
end to
end
what
what
are the
fiat
by fiat
design

choice
structurings
for nested
models
and what
open source
tools
and how
to do
this
and all
this
and then
where
and how
to identify
those
nesting
boundaries
and understand
again
where is
their
like
true
planning
if such
a thing
exists
in the
sense
of
actually
like a
tree
search
let's
just

let's
call that
classic
planning
tree
rollout
then you
get into
the preferential
pruning
and different
attention
focus to
different
specific
potentialities
and there's
a whole
alternative
that's like
the instructionist
and then
that was
contrasted
in the
skillful
performance
paper
and
discussions
with the
interactionist
where
you may
have the
cognitive
system
acting

in a
locally
heuristic
way
without
time steps
of
planning
but
that
doesn't
mean that
from the
side
it wouldn't
look
planned
so what
is really
a plan
and then
how does
that relate
to the
way
these
feedback
loops
roll
forward
and then
there's
the sort
of
retrospective
dynamic
causal
modeling

from the
past
and then
the stepping
into the
procedurally
into the
future
but again
the role
of time
and where
is it
audio
like
with these
coming out
nowhere
and like
a wave
versus
the
the
visuality
which puts
it in a
in a
certain
kind of
all-at-once
frame
that still
has
feedback
or by
analogy
or by
isomorphism

mathematically
yet it's
just so
gripping
at least
in this
that
any visual
feedback
still just
doesn't have
that
gripping
audio
feedback
direct
time
weaving
maybe
though if
there was
a visual
generation
task
like
draw
this
shape
or try
to do
a blind
contour
drawing
then
the visual
feedback
would
stymie

that
and
with
VR
many
of these
things
are being
tested
like
moving
your
eyes
with
a
saccade
or
with
your
head
is
let's
just calibrate
it to
zero
lag
in the
natural
embodied
and
then
what
delay
after
clicking
of a
button
or

moving
ahead
does
it
percolate
back
and
then
how
did
the
timings
and
the
differentials
on
these
cyber
physical
feedback
loops
waiting
for the
next
LLM
token
to
trickle
out
or
waiting
for
someone
to
say
something
in
live

chat
those
informationally
return
to us
like
surfacing
a
journal
entry
from
a
year
ago
are
all
these
different
regular
and
sporadic
or
infrequent
and
sort
of
like
n
equals
one
of
every
unique
event
and
encounter
never
step

in
the
same
river
twice
all
of
that
with
these
regularities
and
then
how
much
there
is
to
extract
informationally
from
even
a
REMA
like
time
lag
generalized
linear
model
with
an
autoregression
a lot
of
signal
is in
that

or
the
Kalman
filter
Bayesian
equivalent
and so
the
emergence
of
these
statistical
extensions
and
computers
directly
like
in
the
data
science
case
or
just
in
the
video
streaming
case
which
is
underpinned
by
so
many
similar
statistical
video

compression
etc
procedures
too
just
abstracted
away
to the
point
where
it can
be like
well
I'm
just
talking
with
you
or
live
streaming
but
that's
on
solid
ground
of
the
composability
of
all
of
these
feedback
loops
within
the
software

and
how
the
concurrency
of
that
whole
architecture
is
orchestrated
even
without
agentic
AI
in
2025
just
from
microservices
or just
a computer
science
perspective
well it
occurs to
me
as we
talked about
destructive
feedback
or disorienting
feedback
and I
see the
examples
in the
chat
of

nausea
seasickness
which is
certainly
disorienting
type of
visual
feedback
but
there's
a
highly
generative
type
I would
say
highly
constructive
type of
visual
feedback
which is
young
children
looking in
the
mirror
and
making
the
connection
that
the
person
in
the
mirror
is

an
image
of
them
and
that
type of
visual
feedback
which is
partially
preceded by
face
recognition
of the
mother
you know
that's
that's a
very positive
feedback
I don't
think we
could
become
I don't
think
intelligence
is
possible
without
feedback
and the
right type
of
feedback
that's
why

educational
institutions
are effective
when they
are
is they
provide a
way for
lots of
different
levels
and
mechanisms
of
feedback
on
what
and how
one is
thinking
about a
given
topic
and
in the
absence of
any type
of feedback
we tend
to
it tends
to be
difficult
so
there's
the opposite
extreme
which is

like the
algorithmization
of that
feedback
where
something like
the YouTube
algorithm
is
directing
and gating
attention
based on
its
mechanisms
and a
video going
viral is an
effect of
the reflexive
feedback loop
that just
amplifies
in the
social
network
but that
can also
be very
destructive
lots of
people go
viral for
reasons that
they don't
want to
and that's
how mob

justice occurs
it's the
same type
of amplifying
feedback loop
so by
analogy
everyone has
experienced
this
I was
recently
reading the
book
everything I
need to know
about business
I learned

from the
grateful
dead
and it
talked about
the
importance
of
the
grateful
dead
audience
to the
grateful
dead
performance
that their
recorded albums
were never

at the
standard of
their live
performances
and that the
band really
counted on
an engaged
audience to
give them
an energy
flow
they could
not have
played the
same
to a
they could
not have
played the
same
in the
studio
and that
was how I
felt when I
was teaching
and lecturing
with a live
classroom
there's a
story told
I think it's
apocryphal
of
people change
the name
but of a

great
mathematician
who was
giving a
lecture series
and one
day none
of the
students
showed up
and his
colleague
comes by
and finds
him still
writing on
the board
and talking
about the
equations
and his
colleague
says you
know
professor
x
what are
you doing
and the
professor
says I'm
teaching
and his
colleague
says to
whom
and the
professor

says what
difference does
that make
and so you
can have that
type of open
loop control
where you're
going to say
whatever you're
going to say
but there's
also I think
tremendous power
even in the
sense to felt
understanding of
the audience
you know that's
why it's so
that's why it
helps so much
when there's
something in the
chat or when
you say
something because
otherwise I
don't want to
spend time just
talking for an
hour about my
thoughts with
no expectation
that it's
generating any
other thoughts
for anyone

else that's
one idea
of what
makes a
thought
real is
that a
thought that
is real
goes out
and makes
other thoughts
occur
it makes
other thoughts
occur in
people's
heads
the surrealist
artist Keith
Wickdor gave
me that idea
and I've
always thought
it's a really
powerful definition
real

Well, on the technical part, Octopus, I think, will have a really epistemic and collaborative and productive year,

connecting some of the dots with, do generalized coordinates on a Bayes' graph, as per Bayesian mechanics,

equal generalized dynamical systems?

How are you all are exploring and implementing these questions?

Where in this stack, from the symbolic choices that are used in the approximating and modeling on through the implementations,

it's quite a space.

So that's fun, and I think for future discussions and works,

let's kind of just have a last piece here about where you want it to end up.

Because I think we can pop into the Bayesian statistical and, but pointing to bigger directions,

we'll continue to prompt and give these open learning experiences,

and have feedback be something that's so rich that we continue to act and first serve with, not just a homework problem that we're getting graded on like a true-false.

And keep getting wrong.

Keep getting red X's on.

If you keep getting wrong, red X's on the true-false problem.

I don't know what to do.

Why not structure it by lifelines?

Strupe wrote.

Yeah, I'm intrigued.

I was intrigued by that question and wasn't sure what lifelines meant in that context.

I assumed it was about structuring the due dates of assignments.

Oh, like, is it like in Who Wants to Be a Millionaire, where you add more resources,

according to the difficulty of the question, add more access to resources,

like you can call someone or give more hints or things like that.

Is that what is meant by lifelines?

We will see in the chat also, it makes me think of the trace that a ball on Waddington's epigenetic landscape,

and this energy landscape-like, ball optimization, ball rolling downhill,

mechanical gradient descent, Newtonian mechanics, Bayesian mechanics,

kind of this nomadic concept,

and the field-like or distributional nature of this channel,

be it narrow or wide,

and the wobbles through procedural time,

or through real embodied time,

and then there's the slice,

and this would give you a Gaussian distribution,

like censored here,

just focusing on that one part,

or a mixture of Gaussians, or whatever it happened to be.

And that sort of statistical ensemble summary statistic,

statistics, modeling,

here we can summarize this channel with its mean and variance,

as can, you know, coin flip with this distribution,

and then the particularity and the historiosity of the ball's lifeline, trace,

pheromone trail, journey,

actual, you know,

where it actually is at any given time.

In contrast with the sort of all of the lateral and antecedent and postcedent ghosts or alternatives,

that it could actually be enabled or constrained to do,

like situational, actual adjacencies,

or just from a third-party modeling perspective,
which could make a great ultra-efficient approximator for a complex thing,
like a lot of factors influence height,
but you still might be able to summarize it with the mean and the variance for a given adequacy,
and just like every other map territory,
richness,
and again, like where's feedback and all that?
Well, one thing that I think would be
cool to do,
I've kind of hit on the idea that broadly speaking,
I'm interested in information processing systems,
which is so broad that you can make it be anything,
but that includes like management cybernetics
and how it was able to draw
useful analogies between engineering practices
and the design of companies.
It has a lot of implications for DAOs,
decentralized autonomous organizations,
like Alcoholics Anonymous,
and how is the,
there's a decentralized structure to Alcoholics Anonymous.
There's also a belief feedback loop
about alcohol that is interrupted.
Gregory Bateson did that analysis cybernetically,
so there's an interesting connection there.
What I think would be really cool is to begin to develop
in the same way that there's a periodic table of elements,
and we understand how basic elements create molecules,
or in the same way that we know that there are
17 fundamental wallpaper groups
of symmetries of the two-dimensional Euclidean plane,
but to have basic experiential feedback patterns,
and to be able to predict the basic low-level mechanisms,
things like analogous to capacitors, resistors, etc.,
in engineering,
and to predictably create specific feedback patterns,
both in terms of a theoretical knowledge,
where, okay, if I can write it down on paper,

that if I move the camera this way through the screen,
I know what the output will be,
but also in a practical way
where there are people
who can make the patterns reliably
given a certain set of tools.
So right now I have access to three cameras and two monitors.
My phone died, unfortunately.
But that feels like a good playground to start with.
What can we do with three cameras and two screens,
and the different ways that we might arrange them,
the different ways that we might move the cameras through the screens,
the different ways that we might move the screens,
and the different ways that we might put each of these things
in geometric location to each other.
I have a sense that this research is probably out there,
that people have done these experiments.
I have a vague memory of reading in a popular science book
about researchers and some open-minded place
in like California or Santa Fe
who did this and cataloged specific patterns.
So if the research is already out there,
it would just be awesome to find it again.
And if it's not out there,
then we can do it ourselves
and say I am moving the camera this way through this screen.
Here's the effect I'm getting.
As I move the camera faster,
it increases this effect
and go back to exploring the parameter space
through just play.
But play with records
so that later we can go back
and it's not like,
you know, nobody...
Nobody is able to use diphenhydramine
very profitably as a psychedelic
because it has such a strong memory

disinhibiting effect.
So you trip,
lots of interesting things happen,
but you can't remember anything
that happened the next day.
And we don't want to do that.
We want to do these experiments,
but in a recorded way
where we can then analyze it
and come up with useful information.
So I guess that would be my call to action.
And we have 10 minutes where we could start
or we could make this be the focus
of future research
at a different improvisation session
or in some structured way.
I'm incredibly open-minded generally.
Yeah.
I think modeling a couple of these
real settings that we're in,
like I have two computer monitors
and OBS is capturing itself
at 25 frames per second.
Or we have an analog camera
looking at an analog TV
and they're exactly aligned.
And so it just doesn't get
to this limited tractor.
And then is it a stable
or does a little flicker of light
destabilize?
What's the Leoponib exponent?
Does it twist?
Does it nest?
What are those categories of patterns?
And what is their intersectional co-arising
from necessity, sufficiency, identity?
Where are they excluding

or complementing each other?
Game of Life gives us a great grid world.
And as we've alluded to,
has some feedback-y elements,
like the information content
between something,
every repeater, numbers of cycles
or distance in space-time
might be tremendously informative
for information theory,
statistical setting.
And what we truly have at hand
in this experience too
is cognitive modeling,
webcams, live streaming.
What about one picture a day
of a text file
that's sent on the other end
of the sine wave?
just what are the base graphs?
What's the maids?
GDS?
How do different language models
help us interpret
and work with that?
And so many fun
and improvisational
experiences
that relate a lot
to the process
of cognitive modeling overall,
like putting a finger
on a fabric
and then moving it
to actively determine
the roughness.
Those kinds of examples.
Isocade,

so a lot of primary embodied.
And then this is kind of like
the Drosophila
or just a case
that allows for
really compositional
lab experiments,
field experiments.
Lab as a field,
field as a lab,
all of that
with webcams
and the cyberphysical
tools at hand
and open source
and fun
and interesting
and with people's feedback
welcome.
And if I would add
one thing to that,
that,
like everything you said,
and it occurs to me
that if we,
generally speaking,
it's been useful
to begin
with the physical
and then cyberfy it.
So the first,
you know,
PID controllers
were built
after someone
actually watched
how a ship captain
was choosing

to steer the ship.
The first pilot
steering mechanisms
came from watching
how pilots
tried to use their body
to force
the early plane
somewhere.
And I think,
you know,
we have this
tremendous advantage
of being bespoke
meat computers,
which is an advantage
in a lot of ways.
And once we have
those experiences,
then we can
translate them
into something
where the space
can be explored
algorithmically.
So once I've done
the experiment
a few times myself,
I can then say,
well,
here's what's
basically happening
with the visual sensor
and the information
source.
And I can do things
that would not be
possible for me

to do physically.
I can go from
my two monitor,
two camera setup
to an eight monitor,
eight camera setup
that I don't actually have,
but that I can
extrapolate
algorithmically
because I can
reality check
that I can produce
in code
what I am experiencing
in my physical
processing unit.
and that's the advantage
that's the,
you know,
Wolfram
had about 15 years
of computational advances
from Conway.
And so when Wolfram
was exploring
the space of parameters,
he didn't have to
use graph paper
or a primitive
seeming computer.
He could generate
lots of rules at once
and immediately
see what was happening
and decide which ones
were interesting.
So the tightening

of the feedback loop
enabled the exploration
of the parameter space
to occur
much more efficiently,
which led to insight,
like things like
the four,
four classification
of cellular automata
that Wolfram proposed.
If he had to design
and analyze
all the rules
purely by hand,
it probably would
have been more difficult.
Although, I mean,
doing things slowly
by hand
certainly has value,
the ability
to get,
to instantly generate,
instantly test,
and instantly generate
again
is a major
cognitive tool
for exploring
parameter spaces.
And I see already
that you're playing
with the different patterns.
Yeah, this is like
the hotel room
and can you look
into the infinite mirror?

How many layers deep?
What if I change
the video resolution?
What can we,
now that we can
look at this
traveling wave?
what is making it,
can you
seize that
traveling wave
in one
fell swoop?
Like,
the sandwich transaction
on the blockchain?
Or,
which of these?
And there's going to be
so many
awesome
traveling waves.
This looks like
a video game,
doesn't it?
It does.
And then
we can take this
and do what we want
to with it.
Like,
the ability
to sample,
which is a type
of feedback.
It's one with
an extreme time delay,
though.

But sampling
as a type of feedback
is very slow.
But it's still feedback.
It's still playback,
at least.
So,
I won't get too
into the woo here.
We have our own,
I have my own
theories about
whether or not
the universe
is a recording
or a simulation
or things like that.
But,
and we didn't
get on hyperstition
either,
which is another
powerful form
of feedback.
So,
I think there's
a lot more
to do.
And
I'm excited
that I had
this time
to talk to you
and anyone
who tuned in.
my X
account link
is in the

description
of the video.
I
don't want
to get
botted,
but
if you'll
see it there.
So,
if you would,
if you're interested
in this,
a good way
to get in touch
with me
is to
DM me
there
or reach out
publicly,
that's fine too.
And
this was so much fun.
I hope we get
good feedback
on the recording.
I hope so too.
Thank you,
Octopus.
Really looking
forward to
exploring
all these
facets together
and with
others.
So,

peace.

Peace.

And then

here's how I

end the stream

just while

it's also

still happening.

Turn off

the audio.

Audio's off.

Mine's off.

Mine's on.

Yours is off.

So,

then,

even though I

can hear it,

you won't.

Then,

let's see

in this

let's see

in this

let's see

let's see

in this

let's see

let's see

let's see

minute,
let's see
in this

let's see
in this
minute,

in this

let's see

minute,

in this
minute,
let's see
in this

let's see